Graphs of Sine and Cosine Functions

ANSWER KEY

Section 1: Amplitude and Vertical Shift

1 Find the amplitude and midline equation of $y = -4\cos(x) + 3$.

a. **Amplitude** $= 4$**, Midline:** $y = 3$ ✓

b. Amplitude $= -4$, Midline: $y = 3$

c. Amplitude $= 4$, Midline: $y = -3$

d. Amplitude $= 8$, Midline: $y = 3$

For $y = A\cos(x) + D$, amplitude is $|A| = |-4| = 4$, and the midline is the horizontal line $y = D = 3$.

2 What are the maximum and minimum values of $f(x) = 5\sin(x) - 2$?

a. **Maximum** $= 3$**, Minimum** $= -7$ ✓

b. Maximum $= 5$, Minimum $= -5$

c. Maximum $= 7$, Minimum $= -3$

d. Maximum $= 3$, Minimum $= -3$

$\sin(x) \in [-1, 1]$. Maximum is $5(1) - 2 = 3$. Minimum is $5(-1) - 2 = -7$.

3 A wave oscillates between a maximum of 12 and a minimum of 4. Find its amplitude.

a. 4 ✓

b. 8

c. 6

d. 16

Amplitude $= \frac{\text{Max} - \text{Min}}{2} = \frac{12 - 4}{2} = \frac{8}{2} = 4$.

Section 2: Period and Frequency

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4 Find the period of the function $y = 3\sin(4x)$.

a. $\frac{\pi}{2}$ ✓

b. π

c. 2π

d. $\frac{\pi}{4}$

Period $T = \frac{2\pi}{B} = \frac{2\pi}{4} = \frac{\pi}{2}$.

5 Find the frequency (cycles per unit of x) of $y = \cos\left(\frac{\pi}{3}x\right)$.

a. $1/6$ ✓

b. 6

c. $\pi/3$

d. 3

Period $T = \frac{2\pi}{\pi/3} = 6$. Frequency $= \frac{1}{T} = \frac{1}{6}$.

6 Write a sine function with amplitude 3 and period 4π.

a. $y = 3\sin\left(\frac{1}{2}x\right)$ ✓

b. $y = 3\sin(4x)$

c. $y = 3\sin(2x)$

d. $y = 4\sin\left(\frac{1}{3}x\right)$

Period $4\pi = \frac{2\pi}{B} \Rightarrow B = \frac{2\pi}{4\pi} = \frac{1}{2}$. Function: $y = 3\sin\left(\frac{1}{2}x\right)$.

Section 3: Phase Shift (Horizontal Shift)

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7 Find the phase shift of $y = 2\sin\left(3x - \frac{\pi}{2}\right)$.

a. $\frac{\pi}{6}$ **to the right** ✓

b. $\frac{\pi}{2}$ to the right

c. $\frac{\pi}{6}$ to the left

d. $\frac{3\pi}{2}$ to the right

Rewrite as $2\sin\left(3\left(x - \frac{\pi}{6}\right)\right)$. Phase shift $C/B = \frac{\pi/2}{3} = \frac{\pi}{6}$ to the right.

8 At what value of x does the first cycle start for $y = \cos(2x + \pi)$?

a. $-\frac{\pi}{2}$ ✓

b. π

c. $-\pi$

d. $\frac{\pi}{2}$

Set argument to 0: $2x + \pi = 0 \Rightarrow 2x = -\pi \Rightarrow x = -\pi/2$.

9 Which transformation shifts the graph of $y = \cos x$ to become identical to $y = \sin x$?

a. **Shift right by** $\frac{\pi}{2}$ ✓

b. Shift left by $\frac{\pi}{2}$

c. Reflect across the x-axis

d. Shift up by 1

Since $\cos(x - \pi/2) = \sin x$, shifting $y = \cos x$ to the right by $\pi/2$ gives $y = \sin x$.

Section 4: Modeling Periodic Phenomena

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10 The temperature in a greenhouse follows $T(t) = 15\sin\left(\frac{\pi}{12}(t-6)\right) + 65$, where t is hours after midnight. What is the maximum temperature?

a. $80\,^\circ\text{F}$ ✓

b. $65\,^\circ\text{F}$

c. $75\,^\circ\text{F}$

d. $95\,^\circ\text{F}$

Max temperature $= \text{Midline} + \text{Amplitude} = 65 + 15 = 80\,^\circ\text{F}$.

11 In the greenhouse model above, at what time of day (t) does the maximum temperature occur?

a. $t = 12$ **(12:00 PM / Noon)** ✓

b. $t = 6$ (6:00 AM)

c. $t = 18$ (6:00 PM)

d. $t = 24$ (Midnight)

Sine attains its maximum of 1 when the argument is $\pi/2$: $\frac{\pi}{12}(t-6) = \frac{\pi}{2} \Rightarrow t - 6 = 6 \Rightarrow t = 12$ (Noon).

12 A Ferris wheel of radius 30 ft has its lowest point 5 ft off the ground and completes one rotation every 60 seconds. What is the midline height?

a. 35 ft ✓

b. 30 ft

c. 65 ft

d. 40 ft

Lowest point $= 5$ ft, highest point $= 5 + 2(30) = 65$ ft. Midline $= 5 + 30 = 35$ ft.

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